

Mamenchisaurus



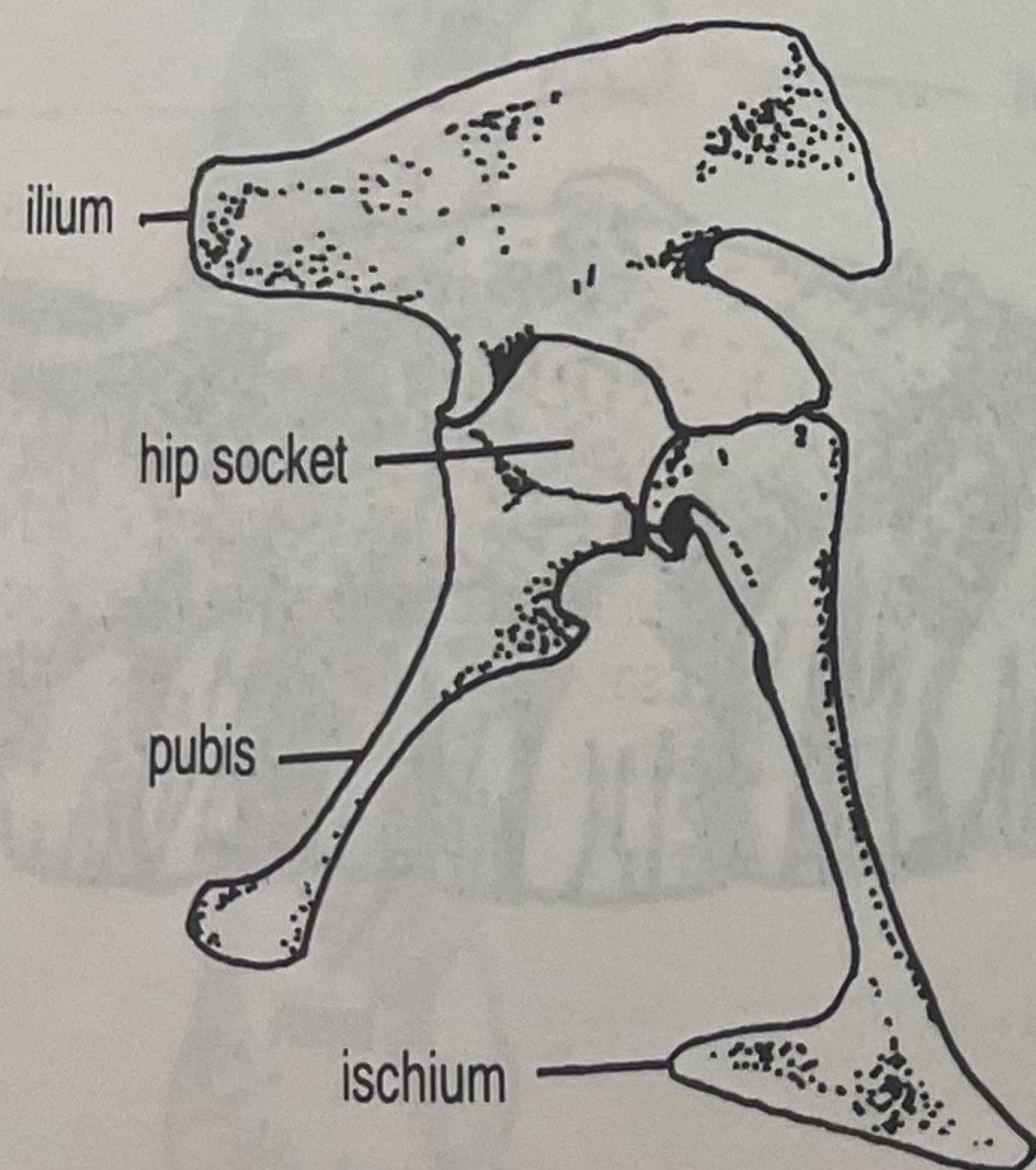
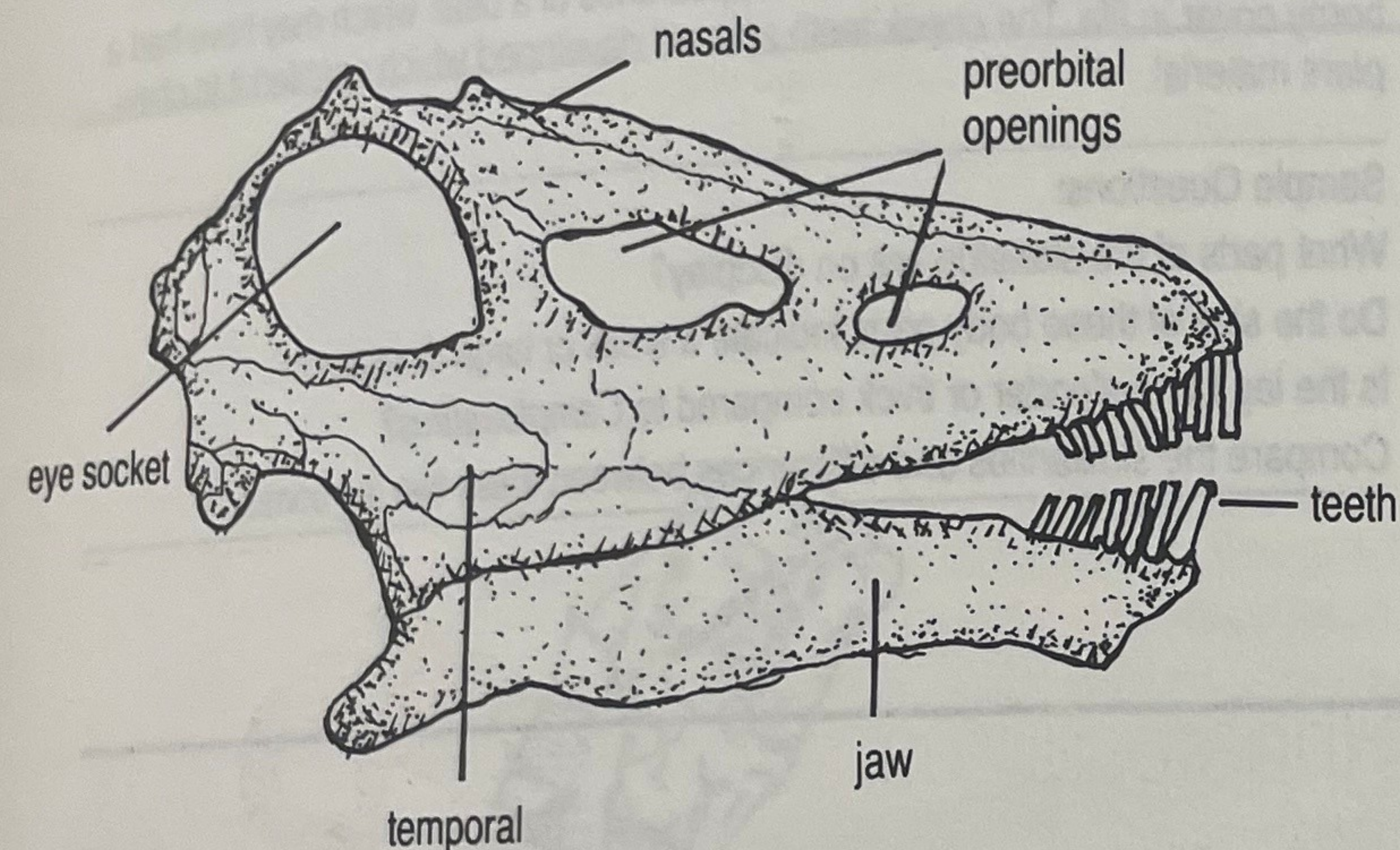
(ma-men-chi-sawr-us)

A quadrupedal plant-eating saurischian ('lizard-hipped') dinosaur (sauropod).

Meaning: 'Reptile from Mamenchi'.

On Display: Reconstruction of near-complete skeleton found in China (Sichuan 'Szechuan').

Size: 22 metres long.



For Discussion: features listed can be easily seen.

Skull: relatively small (with relatively small brain), fairly lightly built (large openings behind the eye sockets: *temporals* and two large openings in front of the eye: *preorbitals*); eye and nasal cavities high up on the skull (may indicate that it was at least partly aquatic, although may suggest it was terrestrial and had a special trunk like a tapir or elephant (with nasals high on the skull); had small peglike teeth in the front of small jaws (indicating a lack of powerful muscles; suggests it ate soft plants, eg. water plants, although some fossil teeth were well worn and suggest it browsed on tough plants, eg. pine needles).

Neck: longer than tail (11 metres - longest of any known sauropod); more than twenty hollowed vertebrae (lightened skeleton) with U-shaped notches (carrying cable-like ligament which provided support) (probably stretched it up to reach tall trees, eg. conifers, pines, firs, to eat food, or lowered it to eat softer plants).

Front Limbs: relatively short and thick with short thick foot bones; five toes (size may indicate bipedal ancestry); straight pillar-like (stood erect like elephant with the sole slightly raised).

Hind Limbs: relatively longer; also very thick and pillar-like (to support body); toes relatively short and thick with large claws (to prevent slipping).

Tail: very long, vertebrae near hipbones with large spines (for large, thick muscles), some fused (rigidity and provided more support); 7 metres long.

Hip Bones: similar to *Gallimimus*, under the hipbone (ilium) socket one bone points down and forward (pubis) and the other points down and backwards (ischium) like lizards.

Sample Questions:

Does it stand on two or four legs? On its whole foot or just its toes?

Is the tail longer or shorter than the neck?

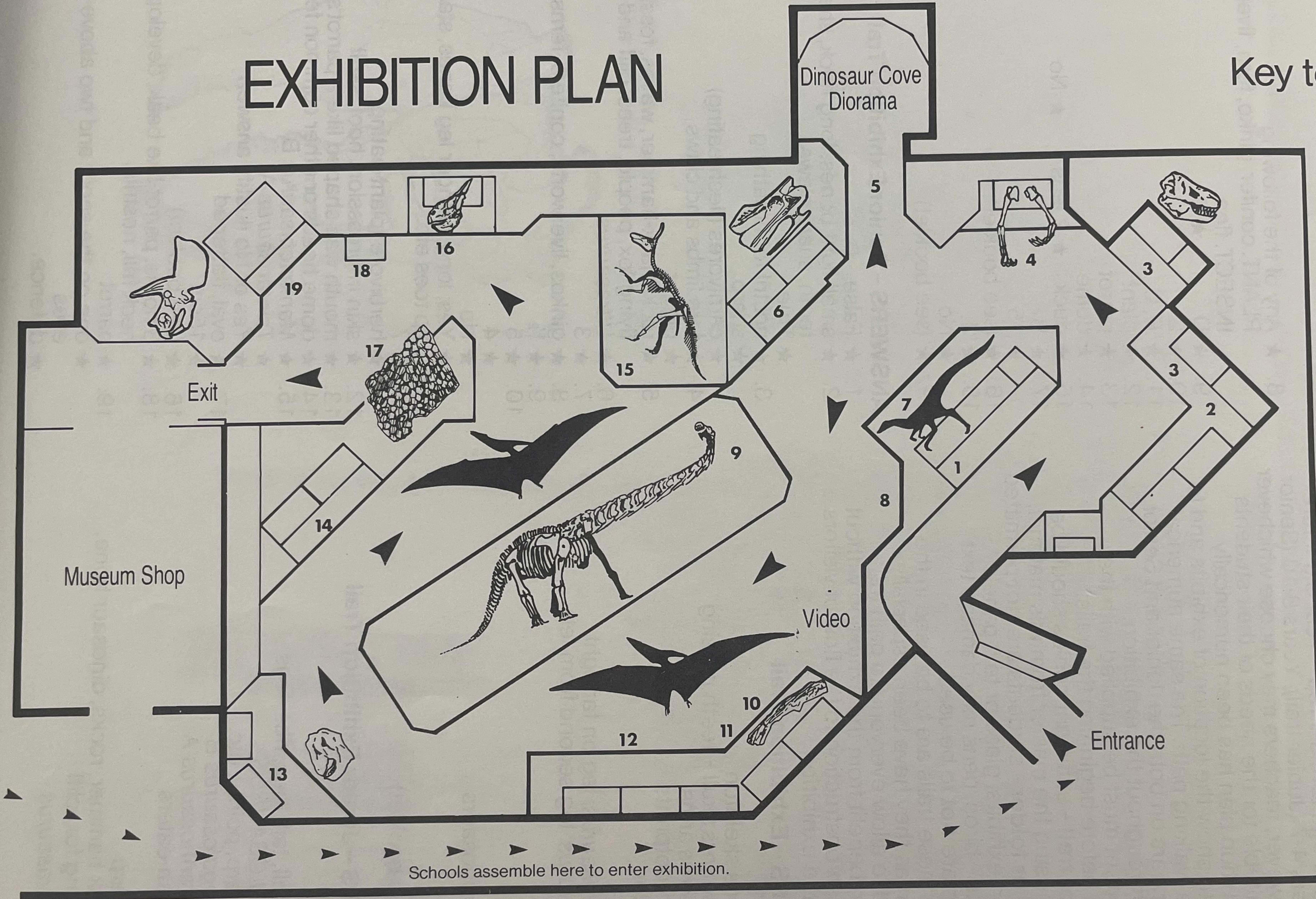
Estimate the lengths of the tail and the neck.

How might the peg-like teeth have been used?

EXHIBITION PLAN

Key to Exhibition Plan

1. *Gallimimus*
2. *Coelophysis*
3. *Albertosaurus*
Megalosaurus
Tyrannosaurus
4. *Deinonychus*
5. Dinosaur Cove Diorama
6. *Rhoetosaurus*
Ultrasaurus
7. Dinosaur footprints
8. Glass Window
9. *Mamenchisaurus*
10. *Hypsilophodon*
11. the Leaellynosaur
12. Hadrosaur
the Lambeosaur
Muttaburrasaurus
Saurolophus
13. *Psittacosaurus*
14. *Gravitholus*
Homalocephale
Pachycephalosaurus
15. *Tsintaosaurus*
17. *Minmi*
16. *Protoceratops*
18. *Triceratops* hatchlings
19. *Triceratops*



Mamenchisaurus hochuanensis (ma-men-chi-sawr-us)

Meaning: Mamenchi is the locality in the People's Republic of China where this kind of dinosaur was first found.

Age: Late Jurassic (140-155 m.yrs.)

Diet: Plant-eater

Size: 22 metres long

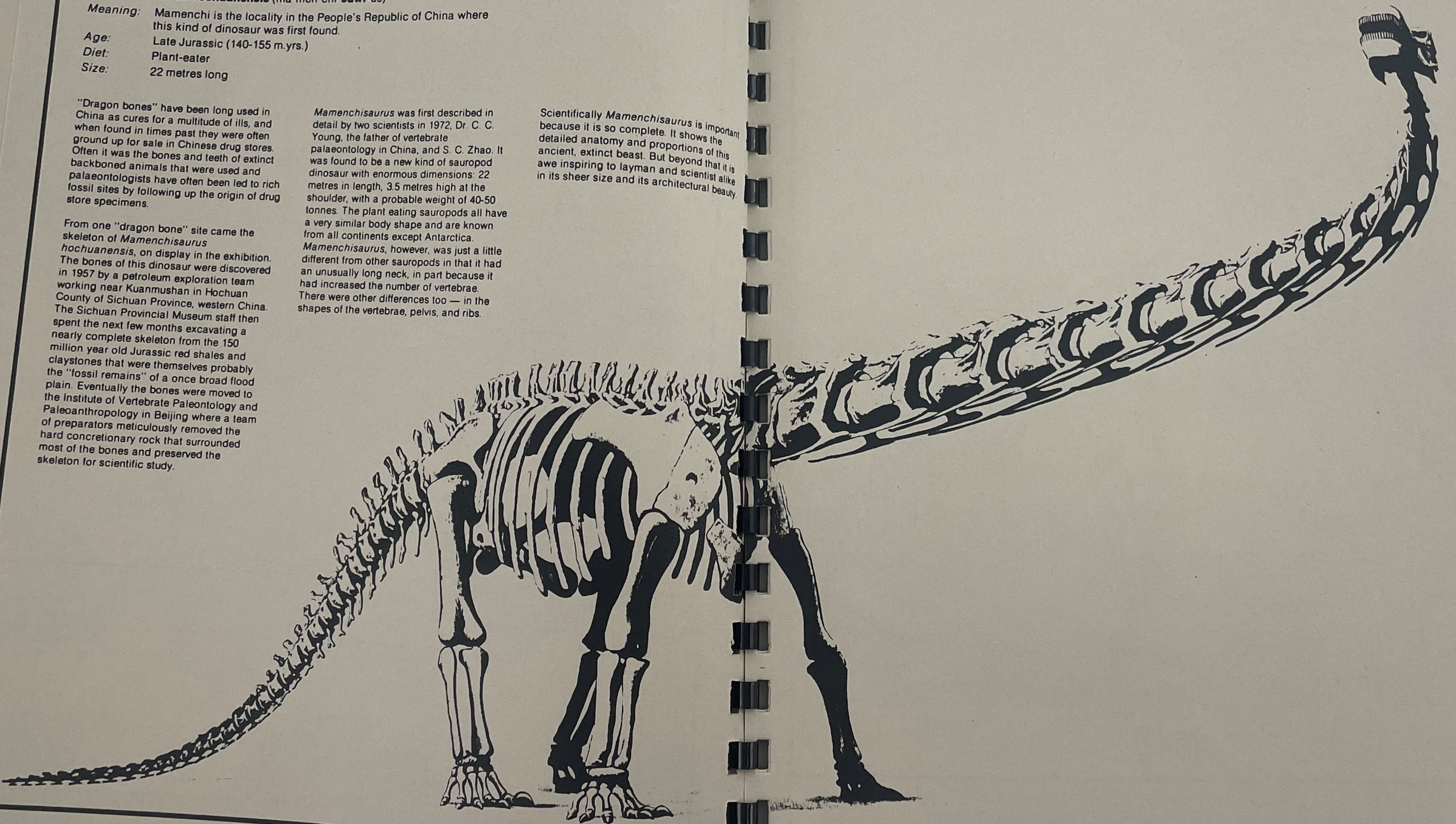
"Dragon bones" have been long used in China as cures for a multitude of ills, and when found in times past they were often ground up for sale in Chinese drug stores. Often it was the bones and teeth of extinct backboned animals that were used and palaeontologists have often been led to rich fossil sites by following up the origin of drug store specimens.

From one "dragon bone" site came the skeleton of *Mamenchisaurus hochuanensis*, on display in the exhibition. The bones of this dinosaur were discovered in 1957 by a petroleum exploration team working near Kuanmushan in Hochuan County of Sichuan Province, western China. The Sichuan Provincial Museum staff then spent the next few months excavating a nearly complete skeleton from the 150 million year old Jurassic red shales and claystones that were themselves probably the "fossil remains" of a once broad flood plain. Eventually the bones were moved to the Institute of Vertebrate Paleontology and Paleoanthropology in Beijing where a team of preparators meticulously removed the hard concretionary rock that surrounded most of the bones and preserved the skeleton for scientific study.

Mamenchisaurus was first described in detail by two scientists in 1972, Dr. C. C. Young, the father of vertebrate palaeontology in China, and S. C. Zhao. It was found to be a new kind of sauropod dinosaur with enormous dimensions: 22 metres in length, 3.5 metres high at the shoulder, with a probable weight of 40-50 tonnes. The plant eating sauropods all have a very similar body shape and are known from all continents except Antarctica.

Mamenchisaurus, however, was just a little different from other sauropods in that it had an unusually long neck, in part because it had increased the number of vertebrae. There were other differences too — in the shapes of the vertebrae, pelvis, and ribs.

Scientifically *Mamenchisaurus* is important because it is so complete. It shows the detailed anatomy and proportions of this ancient, extinct beast. But beyond that it is awe inspiring to layman and scientist alike in its sheer size and its architectural beauty.



Press Release

For Immediate Release

12th August 1987

Dinosaur Display at Auckland Museum

A world-ranking dinosaur display will open at the Auckland Museum in October, in the most ambitious international exhibition the Museum has yet staged.

Called 'Firth's Dinosaurs in New Zealand', the exhibition brings together dinosaur bones and skeletons from China, Australia, North America, Europe and New Zealand which have never been seen before in this country.

"There are some gigantic replicas of dinosaur skeletons from China," said the Director of the Auckland Museum, Stuart Park. "Mamenchisaurus is 22 metres (66 ft) long from head to tail. When it was fully fleshed, it probably weighed between 40 and 50 tonnes. Another, Tsintaosaurus, stands 5.5 metres (17 ft) high."

Firth's Dinosaurs is a co-operative venture between the Museum of Victoria in Melbourne and the Auckland Museum, with support from the Fletcher Challenge company, Firth Concrete Products. Last year, Fletcher Challenge sponsored the Inca Gold Exhibition at the Museum which attracted a then record 67,000 viewers.

"Much of our latest exhibition is currently on display in Melbourne where it is highly successful and attracting huge crowds," said Mr Park. "It is billed there as the best exhibition you will see in a million years and I'm sure that will apply in New Zealand too."

"Although the scale of the Chinese dinosaur skeletons will be the most awesome aspect of the display, the extent of the Australian dinosaur material will probably surprise," added Mr Park.

"Over the last ten years or so, there have been major discoveries of dinosaur material, in Queensland and Victoria in particular," he said. "The Australian continent was rich in small to medium sized dinosaurs including some that seemed to adapt to quite cold conditions."

New Zealand will be represented by its solitary dinosaur relic - part of a tail bone from what is called the Mangahouanga dinosaur. Found in Hawkes Bay, the vertebra was first scientifically described in 1981. Probably about 4 metres (12 ft) long, 2 metres (6 ft) high and weighing 400 kg, New Zealand's only recognized dinosaur to date is believed to have beachcombed the shoreline 65-70 million years ago living on washed-up, huge shellfish called 'ammonites'.

Bones of New Zealand marine reptiles that lived during the age of the dinosaurs, such as the crocodile-like mosasaur, 15 m (45 ft) long, and the plesiosaur, a snake-necked reptile with an overall body length of about 14 m (42 ft) will also be included in the exhibition.

"Among children especially, dinosaurs have always been a source of fascination so we expect the Firth exhibition will really appeal to families," said Mr Park. "Our staff are currently preparing extensive educational kits for schools as well as revising the excellent Australian catalogue which is a library resource in its own right."

He added that although some of the Firth dinosaur exhibition were replicas for practical and security reasons, there was a substantial amount of original material that had never before been seen outside Australia.

"Visitors will actually be able to handle one massive bone that came from a huge dinosaur from the United States," he said. "That's really getting in touch with the prehistoric past!"

New Zealand's first dinosaur exhibition will appear only in Auckland as it is required back in Australia for a January exhibition in Tasmania. "It is a very complex display that takes quite some time to assemble and set up," said Mr Park. "It is insured for \$500,000 and will cost about half that amount to set up and promote in Auckland, hence our thanks to Firth Concrete Products for enabling it all to happen."

For further information, contact Stuart Park (Director) or Sherry Reynolds (Assistant Director) at Auckland Museum (phone (09) 390-443).